Manish CP

Software Engineer - AI & Full-Stack

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Full-stack software engineer with experience architecting AI-driven fintech systems, event-driven AWS backends, and cross-platform mobile apps. Proven track record of owning products end-to-end using Python, TypeScript, React, and React Native.

EXPERIENCE

Founding Software Engineer **Jan 2026 - Present**
Stealth AI Startup - Credit card rewards optimization platform Boston, MA

- Architected the core production API on **Fastify, Prisma, and AWS Lambda**, with **CloudWatch** monitoring and a 285-file test suite gating every deploy. Secured every endpoint with JWT/RBAC, row-level security, and input validation.
- Own the **LLM** enrichment and classification pipeline behind the recommendation engine. A deterministic classifier runs first on every transaction, with the model used only as a fallback for cases it cannot confidently resolve.
- Built the merchant asset system that serves card art and brand colors for thousands of merchants, using **Brandfetch** as the data source. Images are cached through **S3**, **Redis**, and a **CDN**, with automatic fallback when brand data is missing.
- Integrated **Quiltt** (an open banking data provider) for transaction ingestion, with OAuth 2.0 and HMAC-verified webhooks feeding event-driven processing on **AWS SQS, Step Functions, and EventBridge**.
- Led development of the consumer apps across **React Native**, **Next.js**, and **Android/Kotlin**, including **Auth0** onboarding and Google Places integration.
- Independently developed a cross-platform scraping system spanning a **Chrome extension** and native Android client, using a stage-based job pipeline with deduplication.
- Built an internal developer platform covering **52 API endpoints** across 10 functional groups, with user-journey simulation and auth passthrough, so any team member could test the API without prior tooling knowledge.

Software Engineer Co-op **Jan 2025 - Jun 2025**
Berkshire Grey - Autonomous mobile robots for warehouse and fulfillment automation Bedford, MA

- Contributed to real-time data pipelines (**Python AsyncIO, gRPC, Kafka, DB triggers**) serving multiple warehouse systems, fixing bottlenecks that cut end-to-end latency **30%** and brought UI updates to sub-second.
- Wrote **120+** Pytest unit and integration tests with reusable fixtures, wired into Jenkins CI, to catch regressions during the team's **ROS1-to-ROS2 migration** and keep releases stable through the transition.
- Built **React/Redux** interfaces for robotic fulfillment operators, covering four product configurations over REST and WebSocket streams.
- Tested Dockerized workflows across ROS1/ROS2 and Linux simulation environments as part of the team's standard pre-deployment validation process.

Software Engineer - Project Trainee **Mar 2023 - May 2023**
ISRO - Indian Space Research Organization Bengaluru, India

- Built automated **ETL pipelines** in Python processing **10k+** radar records daily, cutting manual preprocessing time by **40%**.
- Developed a space debris tracking prototype in **MATLAB** and tuned orbital coverage algorithms for a **20%** gain in radar pass success rate.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java

Frontend/Mobile: React, React Native, Next.js, Redux, Kotlin

Backend: Django, Fastify, FastAPI, Node.js, NestJS, Spring Boot, GraphQL

Cloud & Infra: AWS (S3, EC2, Step Functions, SQS, Lambda), Docker, PostgreSQL, Redis, Supabase

AI: LangChain, RAG, pgvector, Tool Calling, Agent Design, MCP

Integrations & Auth: Quiltt, Auth0, SendGrid, Google Places API, OAuth 2.0, JWT/RBAC

EDUCATION

Master of Science, Software Engineering Systems - GPA 3.76 **Sep 2023 - Dec 2025**
Northeastern University Boston, MA

Bachelor of Technology, Computer Science and Business Systems **Aug 2019 - May 2023**
Vellore Institute of Technology Vellore, India

PROJECTS

Lease Analyzer Agent | *LangChain, Anthropic API, Ollama, RAG, Flask*

- Built an agentic RAG pipeline with **LangChain** and the Anthropic API to retrieve relevant lease clauses and use tool calling to invoke deterministic analyzers, producing prioritized negotiation briefs instead of relying on free-form generation.
- Engineered a **prompt evaluation** suite with per-case assertions to test retrieval quality, tool selection, and generated outputs, benchmarking a **local Ollama** model against a **frontier** model on the same lease documents.

Car Rental System | *Spring MVC, Hibernate ORM, MySQL, Spring Security, JWT, Thymeleaf*

- Designed a full-stack rental platform on **Spring MVC** and **Hibernate ORM** following a layered architecture (Controller, Service, DAO) with **Spring Security** and **JWT** role-based authentication.
- Resolved booking conflicts under concurrent load using stored procedures, row-level locking, and indexing, with application logic tested via Spring Boot Test and production-aware **MySQL** query optimization.